Mihir Gothal

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EDUCATION

University of British Columbia **Expected Graduation: June 2026**

Bachelor of Applied Science in Mechanical Engineering, cGPA 3.5/4.0

Capstone Project: Development of a Compact and Cost-effective Combustion Chamber for UAV research applications. Designed pre diffuser, liner design calculations and learnt gas turbine combustion processes.

CLUB EXPERIENCE

UBC Formula SAE - Aerodynamics Team | Aerodynamics Team Member **Sep 2023 – Apr 2024**

- Modeled and analyzed the performance of the IC race car's rear wing, using Siemens NX to simulate drag and optimize downforce for competition
- Driver reported greater confidence, with lap times decreasing by over 0.9 seconds due to increased cornering speeds and stability
- IC car reported 5% less tire degradation due to increased downforce

TECHNICAL SKILLS

- CAD/Modeling: Certification in Solid Works, Siemens NX (Aerodynamic modeling), Autodesk Navisworks (3D modeling for oil and gas sites), OpenFOAM and Paraview (CFD course)
- Data & Analysis: Aveva Pi Vision and Asset Framework, SAP, Bluebeam, Power BI, Matlab, Simulink, Gurobi, ANSYS Granta and Microsoft Project
- Fabrication: Design experience in machine shops with manual and CNC lathe, milling machines, drill presses, grinders, waterjet cutter, soft lithography, FDM and SLA
- Relevant Coursework: Computational Fluid Dynamics, Manufacturing Processes, Fluid Mechanics, Heat Transfer Applications, Polymer Engineering, Materials Science and Kinematics of Machinery

WORK EXPERIENCE

Operations Engineering Intern | Pembina Pipelines | Sherwood Park, Alberta **May 2024 - August 2025**

- Created the first pipeline abandonment templates for Pembina at three critical sites including scopes of work, cost estimation, scheduling, screening memos and close out reports
- Supported operations by initiating, coordinating and closing out small projects, utilizing Pembina's management of changes (MOC) process, completing over 30 MOC's including piping modifications, designing new spools and logic changes
- Created the first PD (positive displacement) pump templates in Pembina with flow and volumetric efficiency calculations, pump curves and health monitoring tables using Aveva Pi Vision and Asset Framework, that helped reduce downtime for a major pipeline by over 15%
- Supported documentation, drawing markups, as-builts, validated technical information to be used in process hazard assessments, transient hydraulic modelling and process design
- Learned over 10 standards to assist on over 50 projects for the operations team including CSA Z662, ASME B31.3, ASME 16.5 along with other piping and abandonment standards
- Helped design over 15 safeguards and mitigations from the recommendations presented in HAZOP's
- Created a spike monitoring system in Pi Vision to track when a high alarm was breached filling tanks, resulting in loss of sales. Eliminated issue after monitoring system was signed off, and helped fill 8 major tanks to max volume

TRAINING

- Ground Disturbance Level II – Pembina, 2024
- H2S Alive – Pembina, 2024